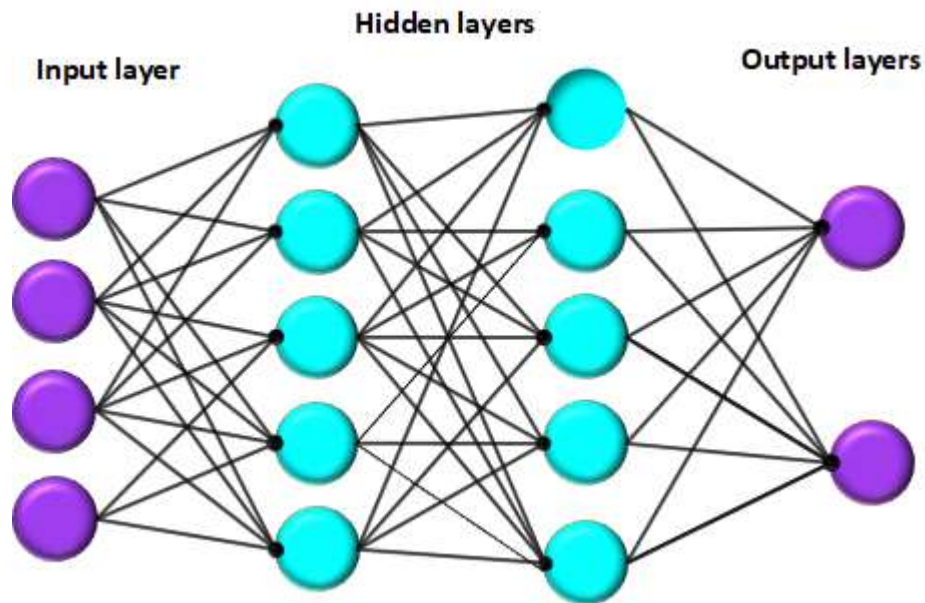


Artificial Neural Network (ANN) from Scratch

An Artificial Neural Network (ANN) is a computational model inspired by the way biological neural networks in the human brain process information. ANNs are a key technology in machine learning and artificial intelligence, used for a variety of applications such as classification, regression, clustering, and more.



ANN Structure

Here is a brief overview of the core components and functioning of an ANN:

Core Components

1. Neurons (Nodes)

- Basic units of an ANN.
- Receive input, process it, and pass the output to the next layer.

1. Layers

- **Input Layer:** The layer that receives the input data.
- **Hidden Layers:** Intermediate layers where the processing is done. There can be one or more hidden layers.
- **Output Layer:** Produces the final output of the network.

2. Weights

- Parameters that are adjusted during training.

- Represent the strength of the connection between neurons.

3. Bias

- An additional parameter in each neuron that allows the model to fit the data better.

4. Activation Function

- A mathematical function is applied to the output of each neuron.
- Introduces non-linearity to the model.
- Common activation functions include Sigmoid, Tanh, ReLU (Rectified Linear Unit), and Softmax.

Functioning of an ANN

1. Forward Propagation

- Input data is fed into the input layer.
- Data is passed through hidden layers, where neurons process it using weights, biases, and activation functions.
- Final output is produced at the output layer.

2. Loss Calculation

- The output is compared to the actual target values using a loss function (e.g., Mean Squared Error for regression, Cross-Entropy for classification).
- The loss function measures how far the predicted output is from the actual target.

3. Backward Propagation

- The error is propagated back through the network.
- Weights and biases are adjusted to minimize the loss using optimization algorithms like Gradient Descent.

4. Training

- The processes of forward propagation, loss calculation, and backward propagation are repeated iteratively.
- The network learns to make better predictions by adjusting the weights and biases.

Example of an ANN Architecture

Example code to build a simple ANN using a popular library like TensorFlow/Keras

```
import tensorflow as tf  
from tensorflow.keras.models import Sequential  
from tensorflow.keras.layers import Dense
```

Create a Sequential model

```
model = Sequential()
```

Add an input layer with 10 neurons (assuming input features are 10)

```
model.add(Dense(10, input_shape=(10,), activation='relu'))
```

Add a hidden layer with 5 neurons

```
model.add(Dense(5, activation='relu'))
```

Add an output layer with 1 neuron (for binary classification)

```
model.add(Dense(1, activation='sigmoid'))
```

Compile the model

```
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])
```

Summary of the model

```
model.summary()
```

Press enter or click to view image in full size

```
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```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 10)	110
dense_1 (Dense)	(None, 5)	55
dense_2 (Dense)	(None, 1)	6
Total params: 171		
Trainable params: 171		
Non-trainable params: 0		

ANN model with Output

Applications of ANN

- **Image Recognition:** Used in tasks like facial recognition, object detection, and image classification.
- **Natural Language Processing (NLP):** Powers applications like sentiment analysis, language translation, and speech recognition.
- **Recommendation Systems:** Used by companies like Netflix and Amazon to recommend products and content.
- **Finance:** Applied in fraud detection, stock market prediction, and risk management.
- **Healthcare:** Assists in disease prediction, medical image analysis, and personalized treatment recommendations.